ROS2 기초편

#25 두산 로보틱스

설치

CONTENTS

01 Docker 설치

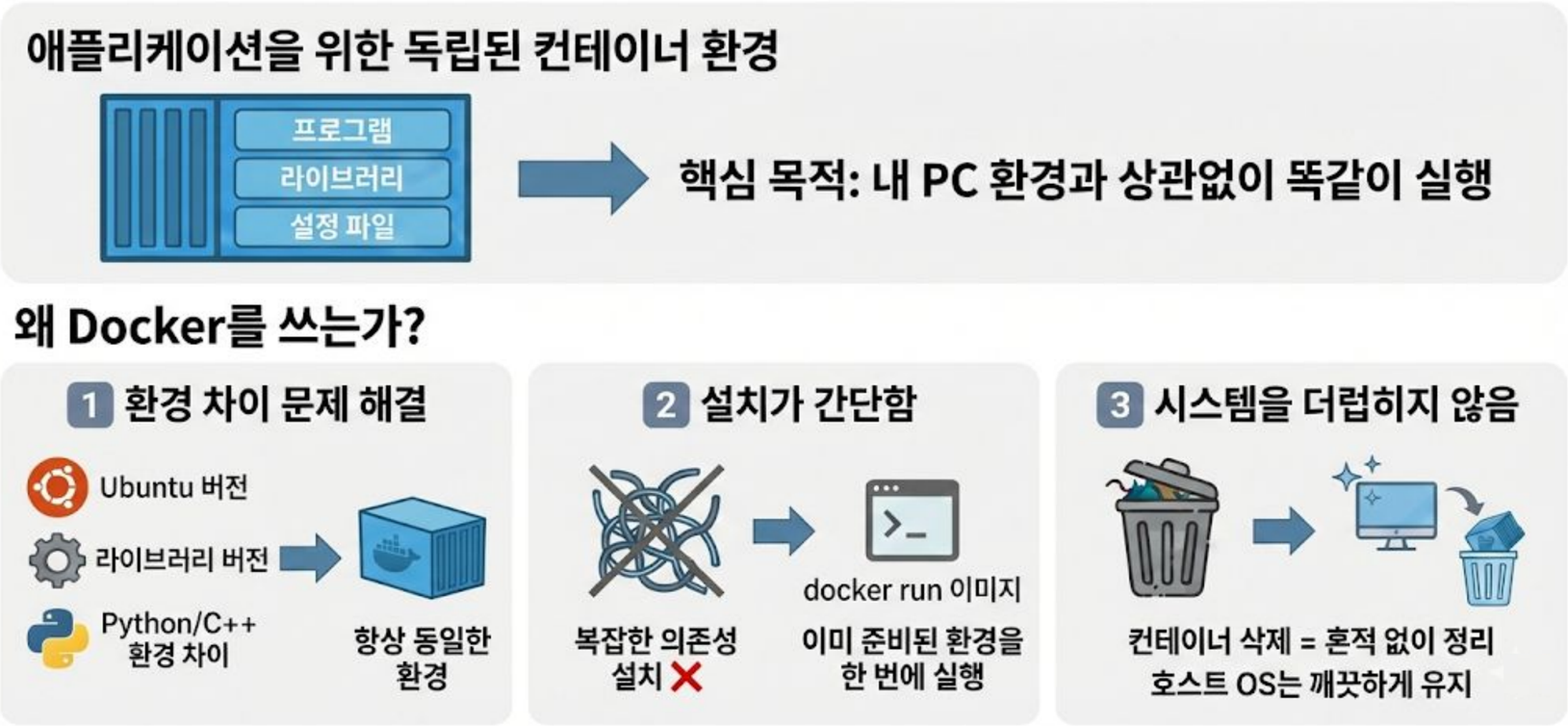
02 두산 로보틱스 설치

03 Moveit2 설치

04 두산 로보틱스 Moveit2 실행

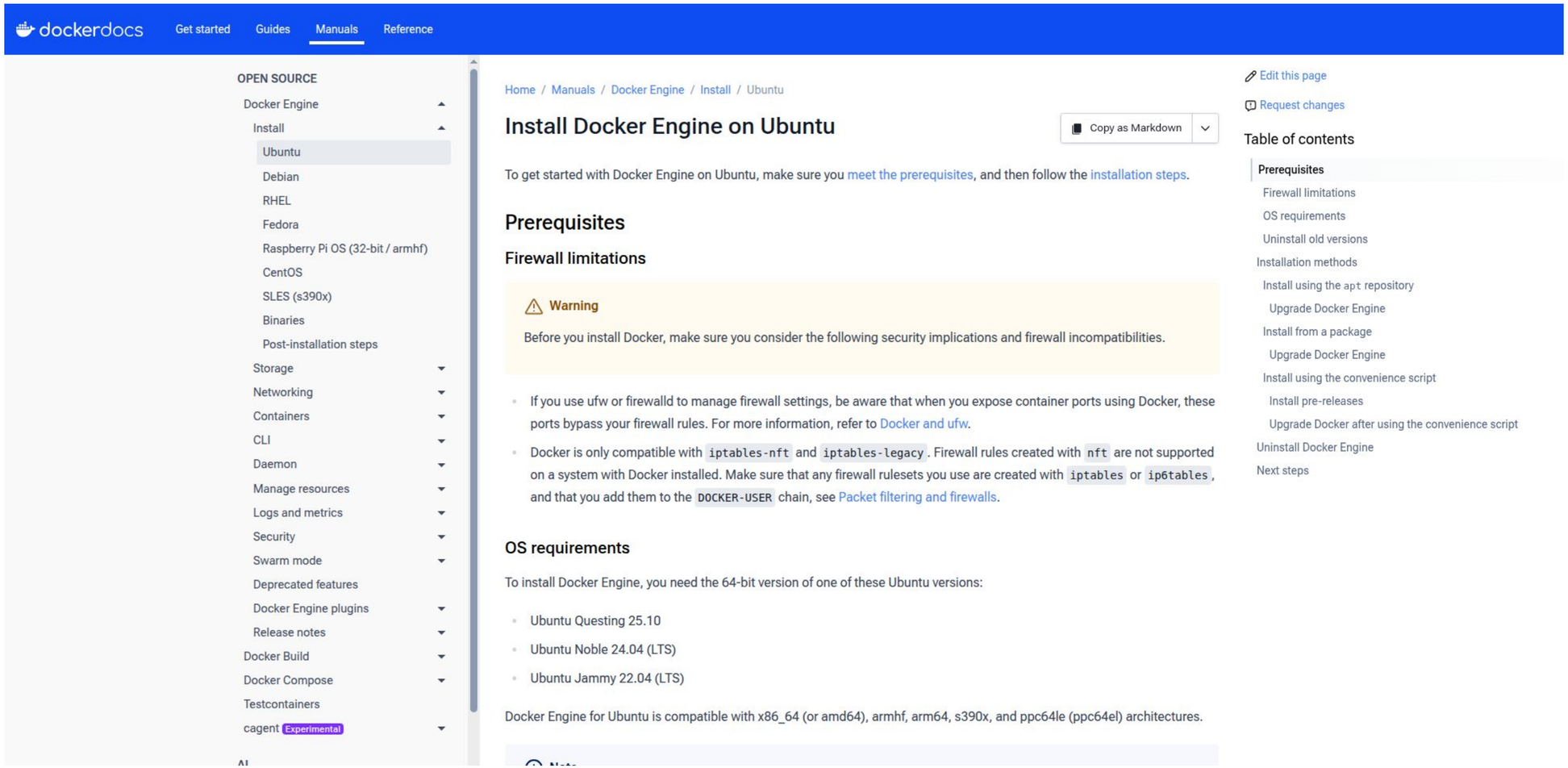
Docker란?

- 프로그램 + 라이브러리 + 설정을 하나로 묶은 '컨테이너'
- 내 PC 환경이 어떻든 어디서나 똑같이 실행



Docker 설치 페이지 접속

- Docker 설치 페이지 : https://docs.docker.com/engine/install/ubuntu/#install-from-a-package



기존 Docker 관련 패키지 제거

- Ubuntu에 예전에 설치된 Docker가 있으면 충돌 방지를 위해 제거
- 명령어 우측 상단의 버튼 누르면 복사 가능

```
$ sudo apt remove $(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-doc podman
```

cop

```
deeptree@robotdive:~$ sudo apt remove $(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-d
oc podman-docker containerd runc | cut -f1)
dpkg: no packages found matching docker-compose
dpkg: no packages found matching docker-compose-v2
dpkg: no packages found matching docker-doc
dpkg: no packages found matching podman-docker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 295 not upgraded.
```

Docker의 apt 저장소 설정

1. Set up Docker's `apt` repository.

```
# Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update
```

```
deeptree@robotdive:~$ # Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update
```

Docker 패키지 설치

2. Install the Docker packages.

Latest Specific version

To install the latest version, run:

```
$ sudo apt install docker-ce docker-ce-cli containe
```

```
deeptree@robotdive:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

Docker가 실행 중인지 확인

- Docker 서비스는 설치 후 자동으로 시작됨, Docker가 실행 중인지 확인하려면 다음 명령어를 사용

```
$ sudo systemctl status docker
```

```
deeptree@robotdive:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-02-05 10:23:54 KST; 1h 3min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 14050 (dockerd)
      Tasks: 18
     Memory: 29.5M
        CPU: 1.316s
     CGroup: /system.slice/docker.service
             └─14050 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 05 10:23:54 robotdive dockerd[14050]: time="2026-02-05T10:23:54.276822925+09:00" level=info msg="Initializin>
Feb 05 10:23:54 robotdive dockerd[14050]: time="2026-02-05T10:23:54.284235221+09:00" level=info msg="Completed b>
Feb 05 10:23:54 robotdive dockerd[14050]: time="2026-02-05T10:23:54.290047419+09:00" level=info msg="Daemon has >
Feb 05 10:23:54 robotdive dockerd[14050]: time="2026-02-05T10:23:54.290102012+09:00" level=info msg="API listen >
Feb 05 10:23:54 robotdive systemd[1]: Started Docker Application Container Engine.
Feb 05 10:31:22 robotdive dockerd[14050]: time="2026-02-05T10:31:22.513308311+09:00" level=info msg="image pulle>
Feb 05 10:31:22 robotdive dockerd[14050]: time="2026-02-05T10:31:22.679779725+09:00" level=info msg="sbJoin: gwe>
```

- Status가 active로 표시되지 않는다면

```
$ sudo systemctl start docker
```

설치 확인

- hello-world 이미지를 실행하여 설치가 성공적으로 완료되었는지 확인

3. Verify that the installation is successful by running the `hello-world` image:

```
$ sudo docker run hello-world
```

```
deeptree@robotdive:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

docker 권한 설정 확인

- 현재 유저가 docker 그룹에 들어가 있는지 확인

```
deeptree@robotdive:~/ros2_ws$ groups
deeptree adm cdrom sudo dip plugdev lpadmin lxd sambashare
```

docker 그룹에 현재 유저 추가

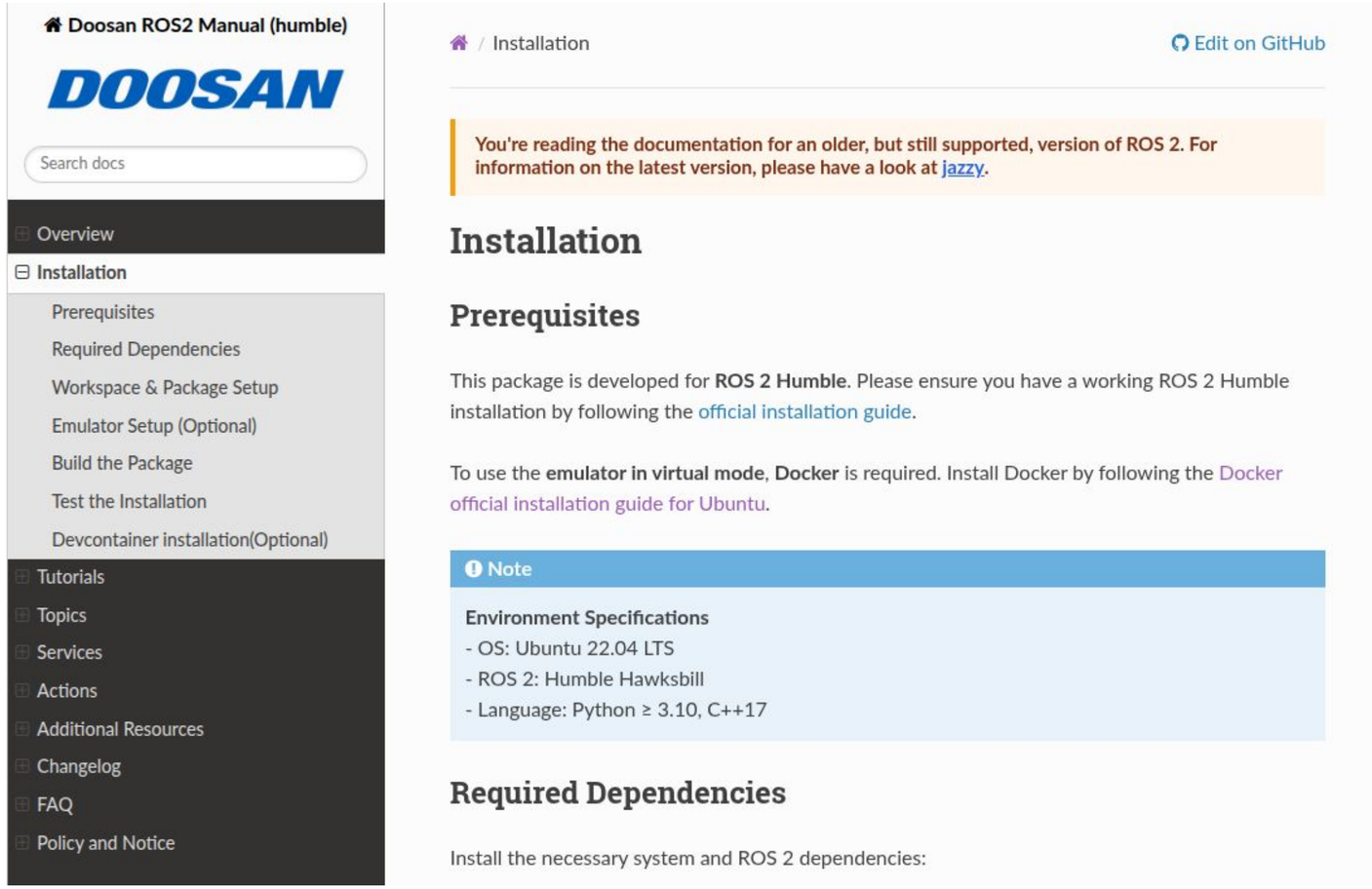
- 유저 추가 했으면 재부팅 후 다시 그룹에 docker가 있는지 확인

```
deeptree@robotdive:~/ros2_ws$ sudo usermod -aG docker $USER
[sudo] password for deeptree:
```

```
ssu@ssu-22663-35:~$ groups
ssu adm cdrom sudo dip plugdev lpadmin lxd sambashare docker
```

Doosan Robotics 설치 페이지 접속

- Doosan Robotics 설치 페이지 :
- https://doosanrobotics.github.io/doosan-robotics-ros-manual/humble/installation.html



필수 의존성 설치

- 필요한 시스템 및 ROS 2 의존성 설치

Required Dependencies

Install the necessary system and ROS 2 dependencies:

```
sudo apt-get update
sudo apt-get install -y libpoco-dev libyaml-cpp-dev wget \
                        ros-humble-control-msgs ros-humble-realtime-tools ros-humble-xacro \
                        ros-humble-joint-state-publisher-gui ros-humble-ros2-control \
                        ros-humble-ros2-controllers ros-humble-gazebo-msgs ros-humble-moveit-ms
                        dbus-x11 ros-humble-moveit-configs-utils ros-humble-moveit-ros-move-grc
                        ros-humble-gazebo-ros-pkgs ros-humble-ros-gz-sim ros-humble-ign-ros2-cc
```

```
deeptree@robotdive:~$ sudo apt-get update
sudo apt-get install -y libpoco-dev libyaml-cpp-dev wget \
                        ros-humble-control-msgs ros-humble-realtime-tools ros-humble-xacro \
                        ros-humble-joint-state-publisher-gui ros-humble-ros2-control \
                        ros-humble-ros2-controllers ros-humble-gazebo-msgs ros-humble-moveit-msgs \
                        dbus-x11 ros-humble-moveit-configs-utils ros-humble-moveit-ros-move-group \
                        ros-humble-gazebo-ros-pkgs ros-humble-ros-gz-sim ros-humble-ign-ros2-control
```

Gazebo 시뮬레이터 지원 설치

- 공식 OSRF 저장소를 추가하고 Gazebo 관련 패키지를 설치

Install Gazebo Simulator support:

```
sudo sh -c 'echo "deb http://packages.osrfoundation.org/gazebo/ubuntu-stable `lsb_release -cs`
wget http://packages.osrfoundation.org/gazebo.key -O - | sudo apt-key add -
sudo apt-get update
sudo apt-get install -y libignition-gazebo6-dev ros-humble-gazebo-ros-pkgs ros-humble-ros-gz-si
```

```
deeptree@robotdive:~$ sudo sh -c 'echo "deb http://packages.osrfoundation.org/gazebo/ubuntu-stable `lsb_release -cs
` main" > /etc/apt/sources.list.d/gazebo-stable.list'
wget http://packages.osrfoundation.org/gazebo.key -O - | sudo apt-key add -
sudo apt-get update
sudo apt-get install -y libignition-gazebo6-dev ros-humble-gazebo-ros-pkgs ros-humble-ros-gz-sim ros-humble-ros-gz
```

워크스페이스 및 패키지 생성

- 워크스페이스를 생성하고 humble 버전으로 git clone

Workspace & Package Setup

Create your workspace and clone the repository:

```
mkdir -p ~/ros2_ws/src
cd ~/ros2_ws/src
git clone -b humble https://github.com/doosan-robotics/doosan-robot2.git
```

```
deeptree@robotdive:~$ mkdir -p ~/ros2_ws/src
cd ~/ros2_ws/src
git clone -b humble https://github.com/doosan-robotics/doosan-robot2.git
```

rosdep init 및 의존성 설치

- 먼저 rosdep 초기화 및 업데이트
- rosdep을 사용하여 패키지 의존성 설치

```
deeptree@robotdive:~/ros2_ws/src$ sudo rosdep init
deeptree@robotdive:~/ros2_ws/src$ rosdep update
```

Install package dependencies using rosdep:

```
cd ~/ros2_ws
rosdep install -r --from-paths src --ignore-src --rosdistro $ROS_DISTRO -y
```

```
deeptree@robotdive:~/ros2_ws/src$ cd ~/ros2_ws
rosdep install -r --from-paths src --ignore-src --rosdistro $ROS_DISTRO -y
```

에뮬레이터 설정

- 실제 로봇 없을 때 에뮬레이터를 이용해 시뮬레이션 할 수 있음

Emulator Setup (Optional)

If you plan to use **emulator mode**, install the virtual DRCF emulator using:

```
cd ~/ros2_ws/src/doosan-robot2
chmod +x ./install_emulator.sh
sudo ./install_emulator.sh
```

```
deeptree@robotdive:~/ros2_ws$ cd ~/ros2_ws/src/doosan-robot2
chmod +x ./install_emulator.sh
sudo ./install_emulator.sh
 pull dsr_emulator version : 3.0.1
 Downloading...
```

패키지 빌드

- 빌드하기 전에 이전 잔여물을 정리
- 워크스페이스 빌드

Build the Package

Before building, clean up previous artifacts (recommended):

```
cd ~/ros2_ws
rm -rf build/ install/ log/
```

Then build the workspace:

```
colcon build
source install/setup.bash
```

```
deeptree@robotdive:~/ros2_ws/src/doosan-robot2$ cd ~/ros2_ws
rm -rf build/ install/ log/
deeptree@robotdive:~/ros2_ws$ colcon build
source install/setup.bash
Starting >>> dsr_common2
Starting >>> dsr_msgs2
Starting >>> dsr_description2
Starting >>> dsr_moveit_config_a0509
Starting >>> dsr_moveit_config_a0912
Starting >>> dsr_moveit_config_e0509
Starting >>> dsr_moveit_config_h2017
Starting >>> dsr_moveit_config_h2515
Starting >>> dsr_moveit_config_m0609
Starting >>> dsr_moveit_config_m0617
```

설치 확인

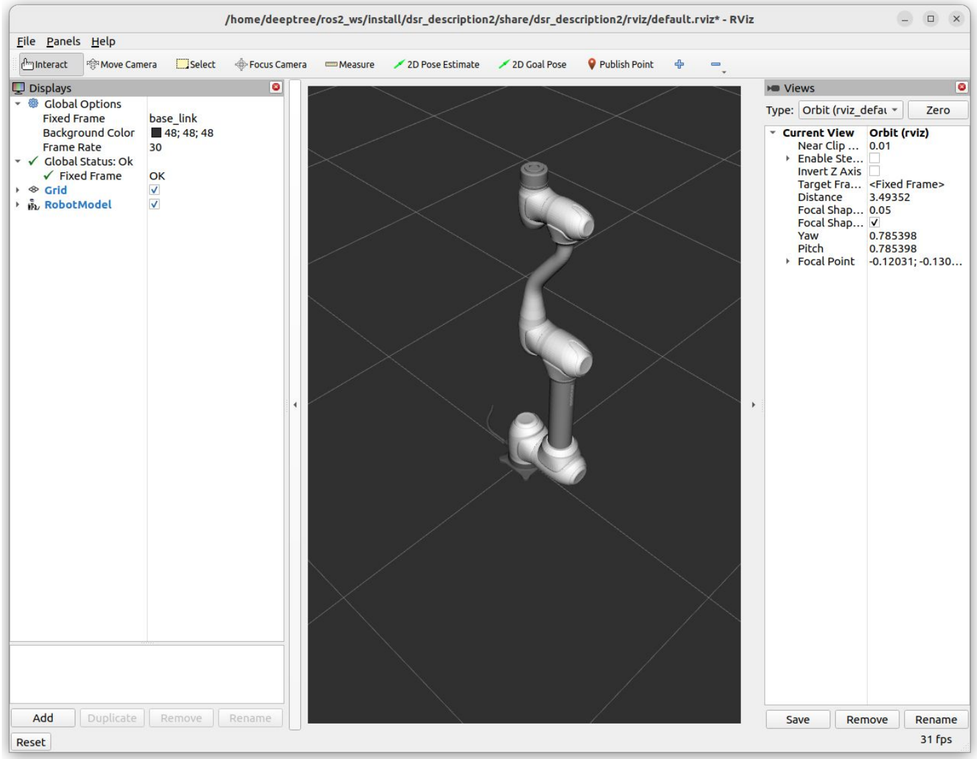
- 설치 확인을 위해 RViz2 데모를 실행
- RViz 화면에 로봇이 잘 생성되는지 확인

Test the Installation

To verify your setup, launch the RViz2 demo:

```
ros2 launch dsr_bringup2 dsr_bringup2_rviz.launch.py
```

```
deeptree@robotdive:~/ros2_ws$ ros2 launch dsr_bringup2 dsr_bringup2_rviz.launch.py
[INFO] [launch]: All log files can be found below /home/deeptree/.ros/log/2026-02-05-14-02-26-042534-r
[INFO] [launch]: Default logging verbosity is set to INFO
[dsr_controller2] Loaded update_rate from YAML: 100
```



Bashrc 명령어 추가

- VS Code 로 .bashrc 수정
- 마지막줄에 alias ros2_ws 명령어 추가
- 저장 후 ros2_ws 실행

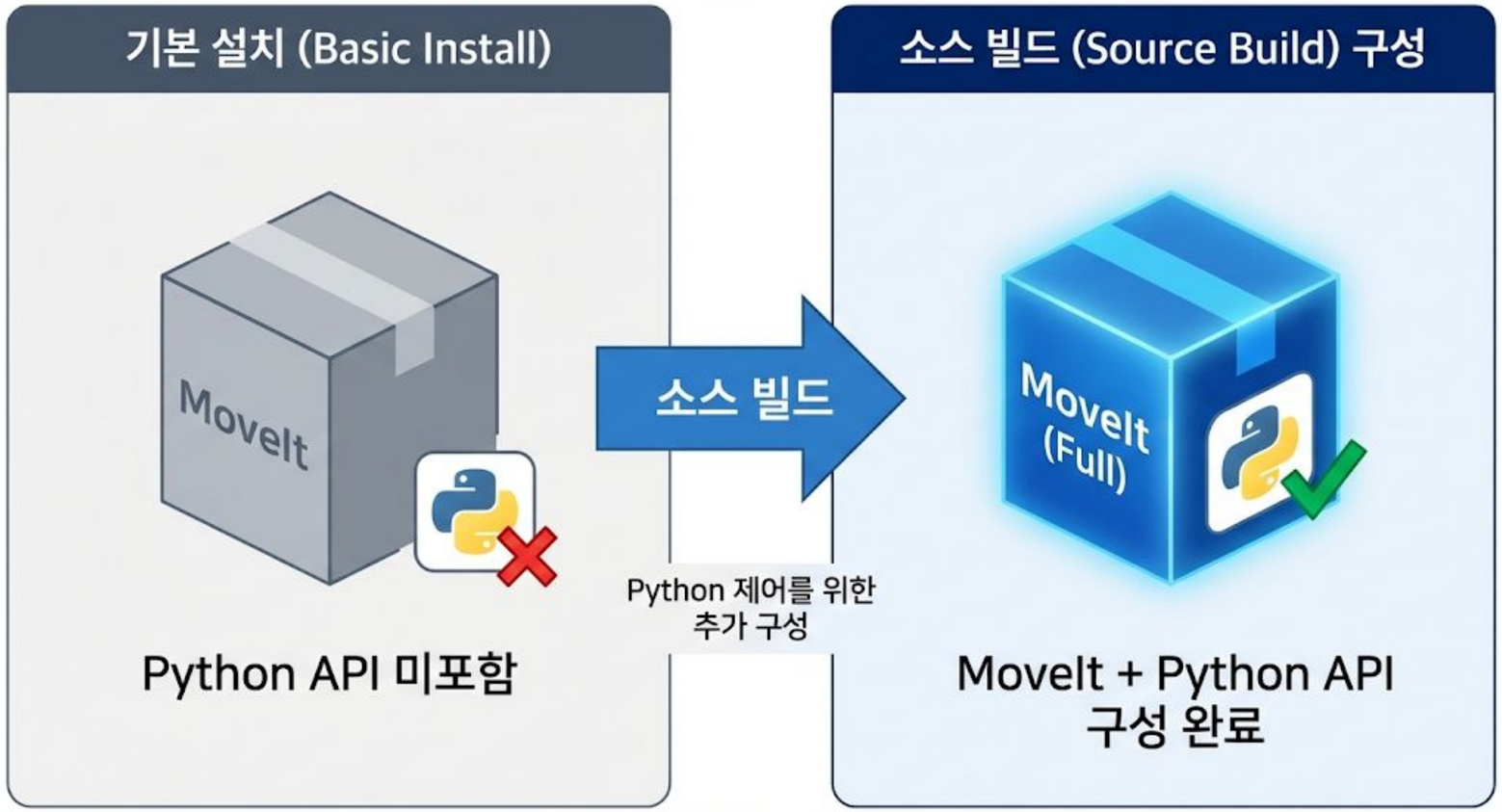
```
deeptree@robotdive:~/ros2_ws$ code ~/.bashrc
```

```
alias sb="source ~/.bashrc; echo \"Bashrc is reloaded!\""
alias ros2_ws="source ~/ros2_ws/install/local_setup.bash; echo \"ros2_ws is activated!\""
```

```
deeptree@robotdive:~/ros2_ws$ sb
Bashrc is reloaded!
deeptree@robotdive:~/ros2_ws$ ros2_ws
ros2_ws is activated!
```

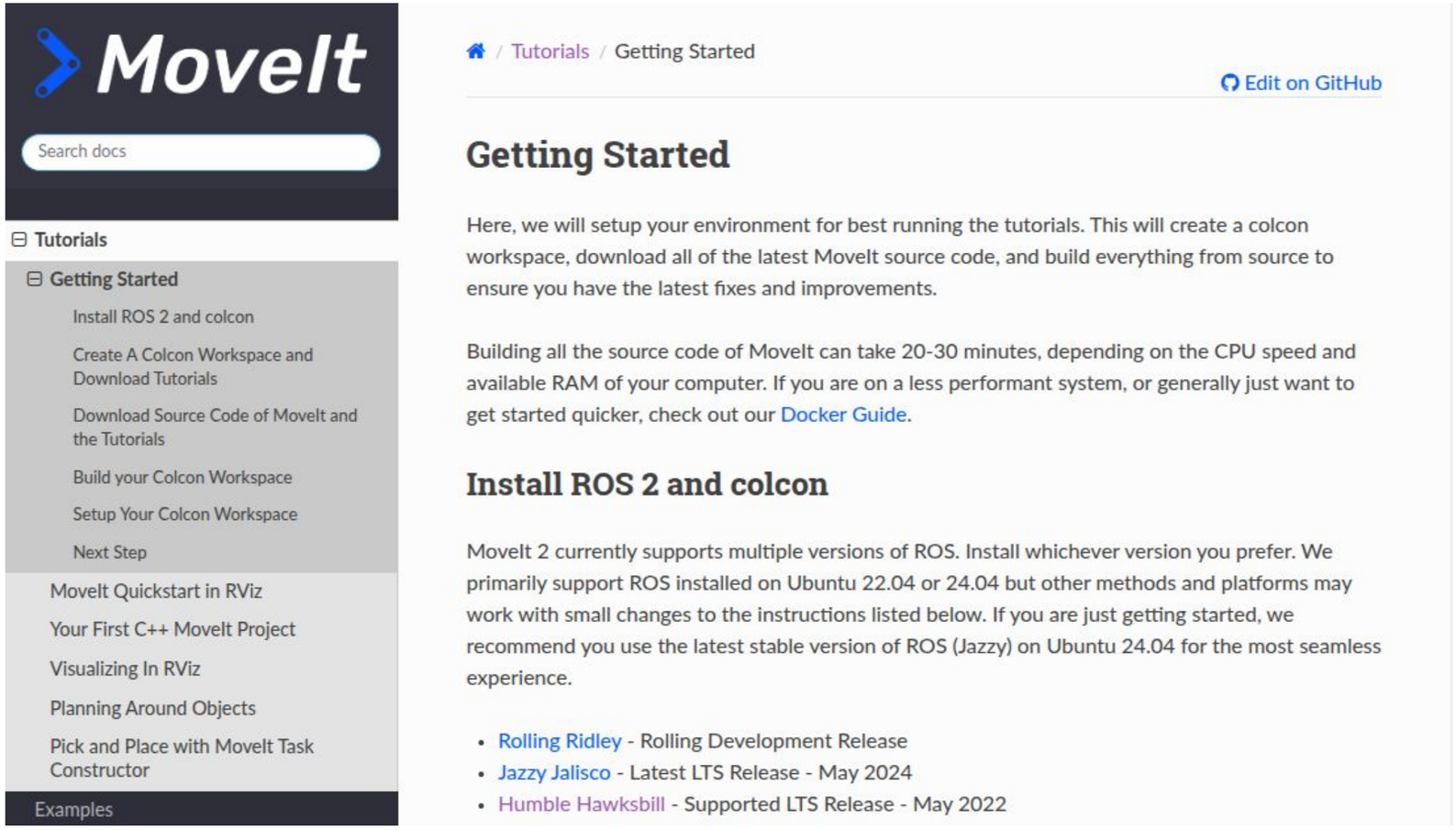
왜 MoveIt을 다시 설치하는가?

- MoveIt은 이미 설치되어 있지만, Python으로 로봇을 제어하기 위해 필요한 MoveIt Python API는 기본 설치에 포함되지 않기 때문에 소스 빌드를 통해 MoveIt을 한 번 더 구성한다.



Moveit2 설치 페이지 접속

- Moveit2 설치 페이지 :
  https://moveit.picknik.ai/main/doc/tutorials/getting_started/getting_started.html



rosdep-python 설치

Install rosdep to install system dependencies :

```
sudo apt install python3-rosdep
```

```
deeptree@robotdive:~$ cd ~/
deeptree@robotdive:~$ sudo apt install python3-rosdep
[sudo] password for deeptree:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-rosdep is already the newest version (0.26.0-1).
python3-rosdep set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 295 not upgraded.
```

rosdep 초기화 및 시스템 최신화

- MoveIt 빌드 전에 시스템을 깨끗한 상태로 맞추는 단계

Once you have ROS 2 installed, make sure you have the most up to date packages:

```
sudo rosdep init
rosdep update
sudo apt update
sudo apt dist-upgrade
```

```
deeptree@robotdive:~$ sudo rosdep init
rosdep update
sudo apt update
sudo apt dist-upgrade
```

colcon 빌드 도구 설치

Install Colcon the ROS 2 build system with mixin:

```
sudo apt install python3-colcon-common-extensions
sudo apt install python3-colcon-mixin
colcon mixin add default https://raw.githubusercontent.com/colcon/colcon-mixin-repository/maste
colcon mixin update default
```

```
deeptree@robotdive:~$ sudo apt install python3-colcon-common-extensions
sudo apt install python3-colcon-mixin
colcon mixin add default https://raw.githubusercontent.com/colcon/colcon-mixin-repository/master/index.yaml
colcon mixin update default
```

vcstool 설치 (여러 저장소 관리 도구)

- vcs는 여러 Git 저장소를 한 번에 관리하는 도구

Install vcstool :

```
sudo apt install python3-vcstool
```

```
deeptree@robotdive:~$ sudo apt install python3-vcstool
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-vcstool is already the newest version (0.3.0-1).
python3-vcstool set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-t
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic linux-too
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Moveit2 워크스페이스 생성

- ws_moveit이라는 MoveIt 전용 워크스페이스 생성

Create A Colcon Workspace and Download Tutorials

For tutorials you will need to have a colcon workspace setup.

```
mkdir -p ~/ws_moveit/src
```

```
deeptree@robotdive:~$ mkdir -p ~/ws_moveit/src
```

MoveIt 튜토리얼 소스 받기

- Clone 할 때 <branch> 부분을 humble로 수정 후 실행

Download Source Code of MoveIt and the Tutorials

Move into your Colcon workspace and pull the MoveIt tutorials source, where `<branch>` can be e.g. `humble` for ROS Humble, or `main` for the latest version of the tutorials :

```
cd ~/ws_moveit/src
git clone -b <branch> https://github.com/moveit/moveit2_tutorials
```

```
deeptree@robotdive:~/ws_moveit/src$ cd ~/ws_moveit/src
git clone -b humble https://github.com/moveit/moveit2_tutorials
Cloning into 'moveit2_tutorials'...
remote: Enumerating objects: 8910, done.
remote: Counting objects: 100% (286/286), done.
remote: Compressing objects: 100% (210/210), done.
remote: Total 8910 (delta 201), reused 76 (delta 76), pack-reused 8624 (from 2)
Receiving objects: 100% (8910/8910), 108.47 MiB | 8.85 MiB/s, done.
Resolving deltas: 100% (5417/5417), done.
```

MoveIt 튜토리얼 소스 받기

- .repos 파일에 정의된 모든 MoveIt 관련 repo를 자동으로 clone

Next we will download the source code for the rest of MoveIt:

```
vcs import --recursive < moveit2_tutorials/moveit2_tutorials.repos
```

```
deeptree@robotdive:~/ws_moveit/src$ vcs import --recursive < moveit2_tutorials/moveit2_tutorials.repos
```

기존 MoveIt 바이너리 제거

- apt 버전 MoveIt과 소스 빌드 MoveIt이 섞이면 충돌 발생

Build your Colcon Workspace

First remove all previously installed moveit binaries:

```
sudo apt remove ros-$ROS_DISTRO-moveit*
```

```
deeptree@robotdive:~/ws_moveit/src$ sudo apt remove ros-$ROS_DISTRO-moveit*
```

MoveIt 의존성 설치 (핵심 단계)

- MoveIt 소스가 필요로 하는 모든 시스템 의존성 설치

The following will install from Debian any package dependencies not already in your workspace.
This is the step that will install MoveIt and all of its dependencies:

```
sudo apt update && rosdep install -r --from-paths . --ignore-src --rosdistro $ROS_DISTRO -y
```

```
deeptree@robotdive:~/ws_moveit/src$ sudo apt update && rosdep install -r --from-paths . --ignore-src --rosdistro $ROS_DISTRO -y
```

MoveIt 빌드

The next command will configure your Colcon workspace:

```
cd ~/ws_moveit
colcon build --mixin release
```

```
deeptree@robotdive:~/ws_moveit/src$ cd ~/ws_moveit
colcon build --mixin release
Starting >>> moveit_resources_panda_description
Starting >>> moveit_common
Starting >>> moveit_resources_pr2_description
Starting >>> srdfdom
Starting >>> launch_param_builder
Starting >>> moveit_resources_fanuc_description
```

- RAM이 16GB 이하라면

```
deeptree@robotdive:~/ws_moveit$ colcon build --executor sequential
```

MoveIt 설치 확인

- Source로 환경을 불러온 뒤 moveit 패키지 리스트 확인

```
deeptree@robotdive:~/ws_moveit$ source install/setup.bash
deeptree@robotdive:~/ws_moveit$ ros2 pkg list | grep moveit
```

```
moveit_chomp_optimizer_adapter
moveit_common
moveit_configs_utils
moveit_core
moveit_hybrid_planning
moveit_kinematics
moveit_msgs
moveit_planners_chomp
moveit_planners_ompl
moveit_plugins
moveit_py
moveit_resources
moveit_resources_fanuc_description
moveit_resources_fanuc_moveit_config
moveit_resources_panda_description
moveit_resources_panda_moveit_config
moveit_resources_pr2_description
moveit_resources_prbt_ikfast_manipulator_plugin
moveit_resources_prbt_moveit_config
```

MoveIt bashrc 추가

```
alias sb="source ~/.bashrc; echo \"Bashrc is reloaded!\""
alias ros2_ws="source ~/ros2_ws/install/local_setup.bash; echo \"ros2_ws is activated!\""
alias ws_moveit="source ~/ws_moveit/install/local_setup.bash; echo \"ws_moveit is activated!\""
```

```
deeptree@robotdive:~/ws_moveit$ sb
Bashrc is reloaded!
deeptree@robotdive:~/ws_moveit$ ws_moveit
ws_moveit is activated!
```

MoveIt bashrc 추가

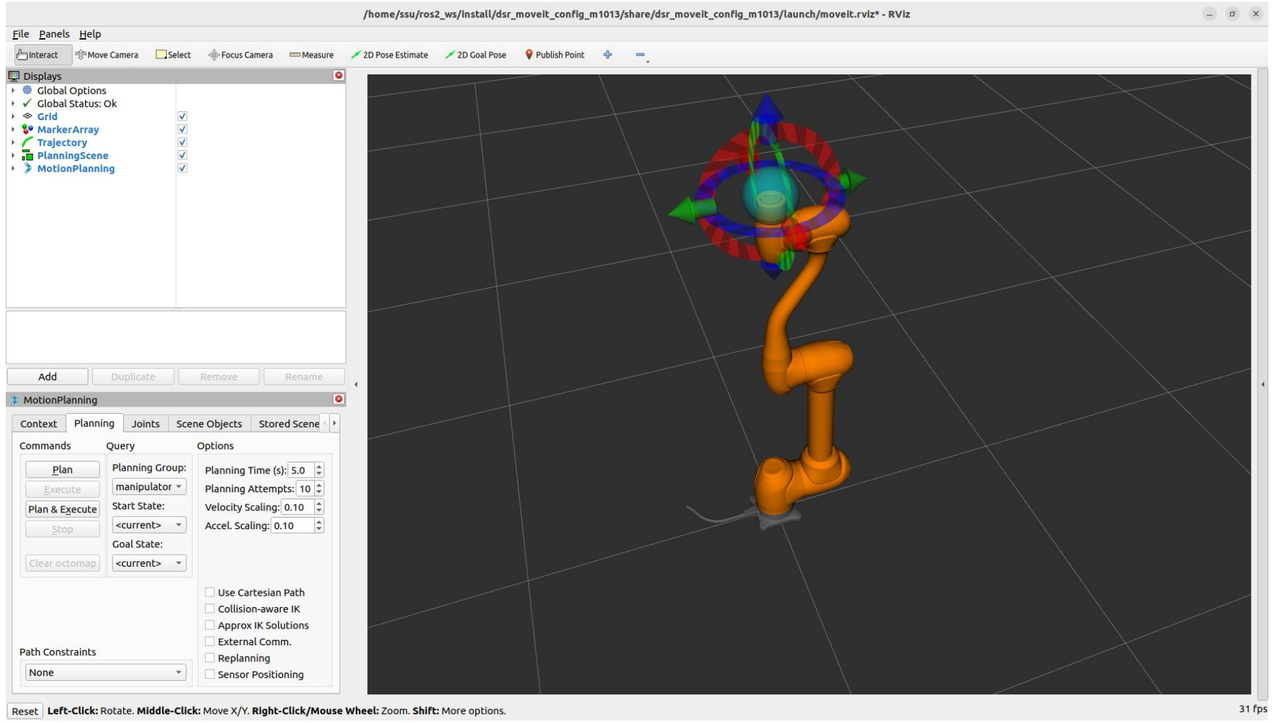
```
alias sb="source ~/.bashrc; echo \"Bashrc is reloaded!\""
alias ros2_ws="source ~/ros2_ws/install/local_setup.bash; echo \"ros2_ws is activated!\""
alias ws_moveit="source ~/ws_moveit/install/local_setup.bash; echo \"ws_moveit is activated!\""
```

```
deeptree@robotdive:~/ws_moveit$ sb
Bashrc is reloaded!
deeptree@robotdive:~/ws_moveit$ ws_moveit
ws_moveit is activated!
```

시뮬레이션 Moveit2 실행

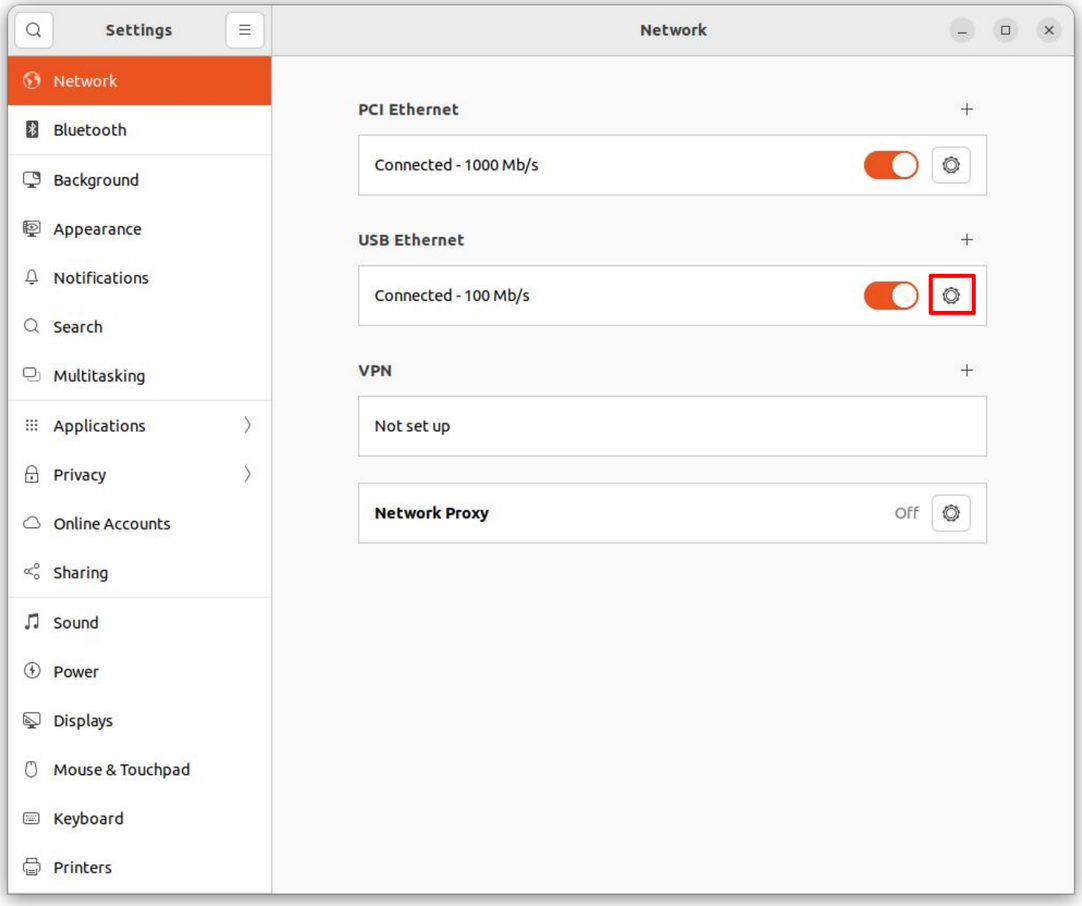
ros2 launch dsr_bringup2 dsr_bringup2_moveit.launch.py mode:=virtual model:=m0609 host:=127.0.0.1

```
ssu@ssu-22663-35:~$ ros2_ws
ssu@ssu-22663-35:~$ ws_moveit
ssu@ssu-22663-35:~$ ros2 launch dsr_bringup2 dsr_bringup2_moveit.launch.py mode:=virtual model:=m0609 host:=127.0.0.1
```



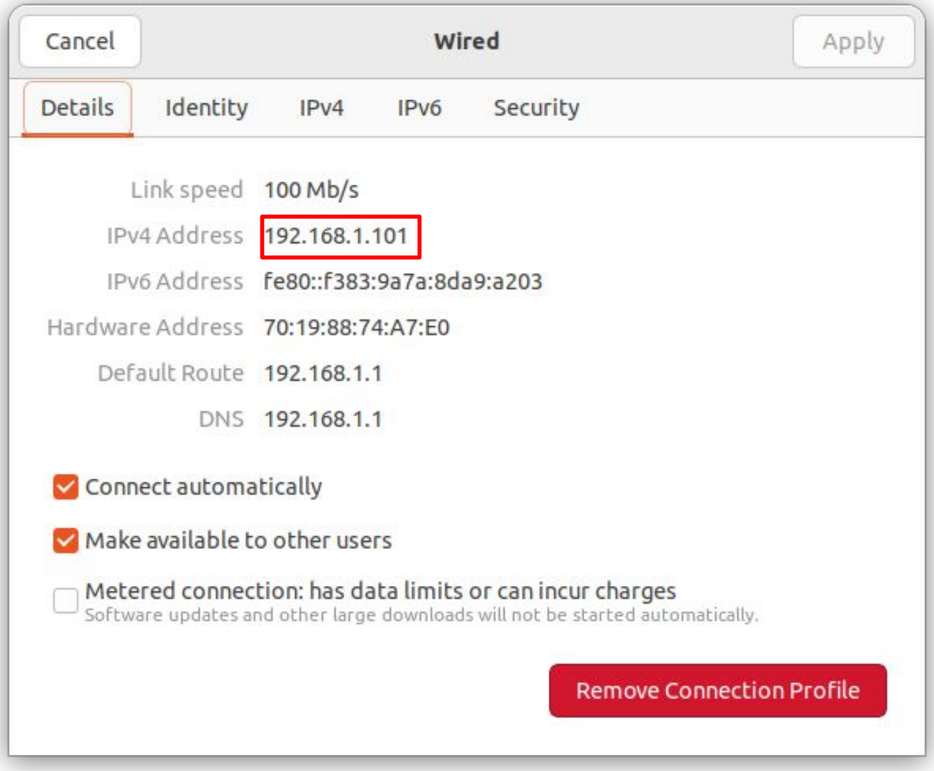
실물 로봇 Moveit2 실행

- 로봇과 PC 간에 이더넷 연결 후 이더넷 설정
- USB 이더넷 설정 클릭



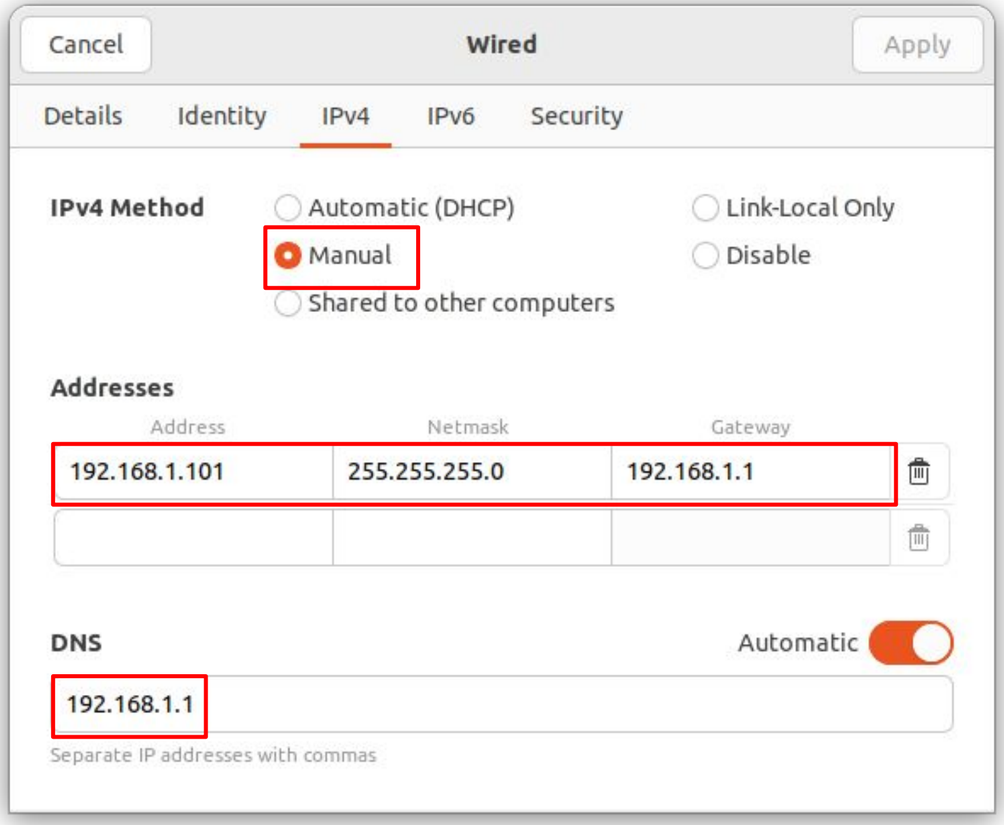
실물 로봇 Moveit2 실행

- IPv4 Address가 192.168.1.xxx 로 되어있는지 확인



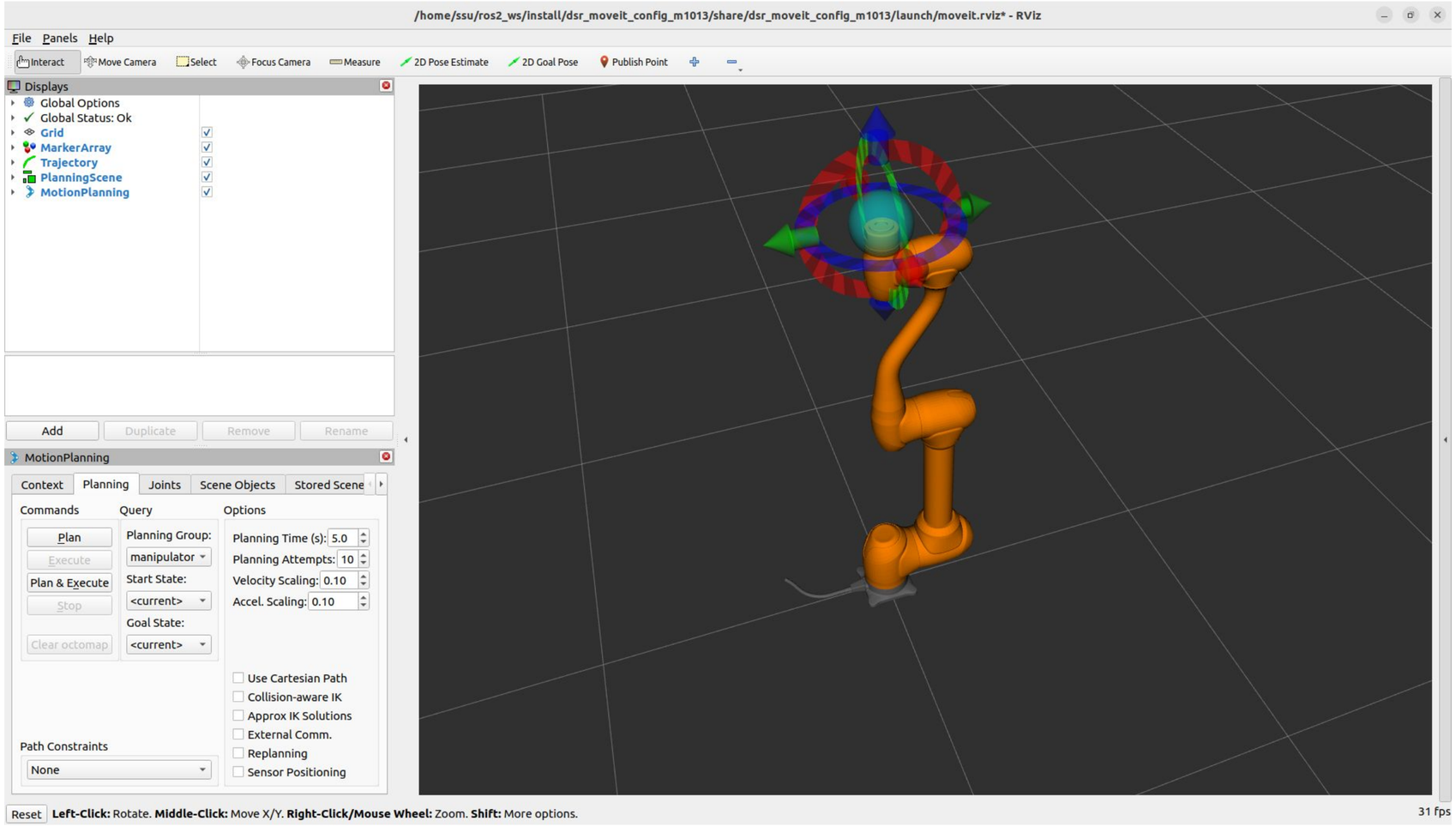
실물 로봇 Moveit2 실행

- 설정돼있지 않다면 IP4에서 IP 주소, DNS 설정



실물 로봇 Moveit2 실행

```
ros2 launch dsr_bringup2 dsr_bringup2_moveit.launch.py mode:=real model:=m0609 host:=192.168.1.100
```

```
ssu@ssu-22663-35:~$ ros2_ws
ssu@ssu-22663-35:~$ ws_moveit
ssu@ssu-22663-35:~$ ros2 launch dsr_bringup2 dsr_bringup2_moveit.launch.py mode:=real model:=m0609 host:=192.168.1.100
```

실물 로봇 Moveit2 실행